16

15 D

$$d\sin\theta = n\lambda \Rightarrow d\sin 28^0 = 2\lambda \Rightarrow \frac{d}{\lambda} = \frac{2}{\sin 28^0}$$

$$d\sin\theta = n\lambda$$

$$\sin\theta = \frac{n\lambda}{d}$$

$$\sin\theta \le 1$$

$$\frac{n\lambda}{d} \le 1$$

$$n \le \frac{d}{\lambda} = \frac{2}{\sin 28^0} = 4.2601$$

$$n_{max} = 4$$

$\therefore$ number of diffraction lines $= (4 \times 2) + 1 = 9$

16 B Distance, $r = 20 = 10^{16}$ m